

BARLAD, Romania

WMO: 151970

Lat: 46.2331N

Lon: 27.6444E

Elev: 569

StdP: 14.40

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.2	12.1	2.9	6.5	10.9	7.5	8.3	14.9	27.6	26.8	22.1	20.9	6.8	0	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	90.2	70.0	87.6	68.9	85.2	68.3	73.1	85.4	71.5	82.6	70.0	81.2	6.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.4	110.2	78.1	67.8	104.4	76.2	66.4	99.2	74.5	37.1	85.5	35.6	82.9	34.3	80.9	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.9	18.4	16.3	2.5	94.1	5.4	3.2	-1.4	96.4	-4.6	98.3	-7.7	100.1	-11.6	102.4		
(5)			1.6	75.7	6.0	1.7	-2.7	77.0	-6.2	78.0	-9.5	78.9	-13.8	80.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.4	28.0	34.3	41.1	52.1	61.8	69.6	71.8	71.5	63.2	51.6	40.6	30.3		
	DBStd	17.29	7.85	9.29	8.58	7.66	6.32	5.02	4.67	5.77	7.60	7.59	8.85	9.05		
	HDD50	2471	681	442	290	67	5	0	0	0	1	72	303	610		
	HDD65	5653	1146	860	740	388	135	16	6	13	124	419	731	1075		
	CDD50	2991	0	2	15	131	370	589	677	668	396	121	22	0		
	CDD65	699	0	0	0	2	36	155	218	216	69	3	0	0		
	CDH74	6568	0	0	2	33	405	1351	1978	2128	650	21	0	0		
	CDH80	2170	0	0	0	3	75	388	686	837	180	1	0	0		
(14) Wind	WSAvg	6.6	7.0	7.3	8.4	7.5	6.4	6.0	5.8	5.6	6.1	5.9	6.4	6.8		
(15) Precipitation	PrecAvg	20.1	0.9	0.8	1.1	1.8	2.3	2.8	2.1	2.0	1.7	2.0	1.3	1.3		
	PrecMax	27.3	1.8	2.7	2.3	3.4	6.2	6.3	5.9	6.2	4.8	7.8	4.1	4.3		
	PrecMin	12.3	0.0	0.0	0.1	0.5	0.2	0.7	0.3	0.5	0.2	0.1	0.0	0.1		
	PrecStd	4.6	0.6	0.6	0.6	0.8	1.4	1.5	1.3	1.3	1.1	1.6	0.9	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.8	61.1	69.2	77.8	86.1	90.3	92.3	94.1	88.1	77.0	66.7	53.5		
		MCWB	43.0	51.9	53.6	61.7	66.9	72.3	71.7	72.1	65.9	63.0	56.6	48.8		
	2%	DB	43.4	54.6	62.1	73.4	81.7	86.8	89.5	91.0	85.1	71.6	61.2	49.1		
		MCWB	40.9	47.9	50.1	58.2	64.9	70.2	71.0	69.3	65.9	59.6	53.9	46.0		
	5%	DB	40.4	49.9	58.0	69.3	78.6	84.5	86.9	88.0	81.4	67.6	56.3	45.4		
		MCWB	38.6	44.8	48.3	56.4	64.2	69.3	69.8	68.0	63.7	57.7	51.3	42.6		
	10%	DB	37.8	46.0	54.0	65.7	75.1	81.9	84.1	84.9	77.4	63.6	52.9	41.0		
		MCWB	36.3	42.1	45.8	54.7	62.0	68.2	68.6	67.3	63.4	55.6	48.9	38.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.3	53.2	54.5	62.7	69.6	74.8	74.8	75.3	70.0	65.1	58.0	50.8		
		MCDB	47.0	61.1	66.4	76.0	83.6	86.2	86.9	90.4	81.7	72.5	63.9	52.1		
	2%	WB	41.3	48.4	51.1	59.7	67.3	72.7	72.9	72.6	68.4	61.2	54.9	46.4		
		MCDB	42.8	53.2	59.7	71.5	79.4	83.6	84.9	85.3	78.5	69.7	60.2	49.3		
	5%	WB	39.2	45.3	48.9	57.5	65.3	71.0	71.4	70.5	66.6	59.0	52.5	42.6		
		MCDB	40.6	48.9	55.8	67.1	75.9	81.2	83.0	82.4	75.4	66.1	56.5	45.4		
	10%	WB	36.7	42.7	46.7	55.6	63.1	69.4	69.8	68.8	64.7	56.7	49.5	39.1		
		MCDB	38.1	45.3	52.3	64.1	72.4	79.6	80.7	80.6	73.2	61.8	52.5	40.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	12.3	15.2	18.0	19.7	19.8	20.0	21.1	19.5	15.9	11.1	9.5		
		MCDBR	12.4	19.1	22.8	25.0	25.1	23.6	24.5	25.8	25.4	22.2	17.5	13.4		
	5% WB	MCWBR	9.4	12.9	13.5	12.5	11.6	9.7	9.3	9.0	9.9	11.1	9.9	10.6		
		MCWBR	11.1	18.8	20.9	22.4	23.4	21.5	21.0	23.1	21.0	20.0	15.4	13.2		
(40) Clear-Sky Solar Irradiance	taub		0.327	0.358	0.394	0.452	0.423	0.421	0.443	0.454	0.412	0.379	0.353	0.326		
	taud		2.284	2.238	2.198	2.060	2.223	2.264	2.210	2.169	2.249	2.321	2.337	2.312		
	Ebn at Noon		238	253	261	255	268	269	261	252	252	243	226	224		
	Edh at Noon		25	33	39	49	43	42	43	44	37	30	24	23		
(44) All-Sky Solar Radiation	RadAvg		401	649	1027	1407	1798	1892	1912	1735	1242	799	427	325		
	RadStd		54	82	102	132	153	159	149	124	123	90	42	39		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (48 neighbors)		+1.13	N/A	N/A	+1.24	N/A	N/A	-247	-329	+128	+31

Nomenclature: See separate page